

MAGADAN SOKOL, Russian FederationWMO: **259130**Lat: **59.911N**Lon: **150.720E**Elev: **574**StdP: **14.39**Time Zone: **11.00 (E11)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.2	-34.5	-44.9	0.4	-37.7	-40.2	0.6	-33.4	24.4	20.1	21.5	19.2	2.7	50	0.796

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	73.4	59.4	69.7	57.8	65.9	56.0	61.3	70.0	59.2	66.5	57.4	63.6	7.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.5	72.2	63.3	55.8	67.6	60.8	54.6	64.7	59.2	27.6	70.1	26.1	66.4	24.9	63.4	69.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.3	17.0	14.8	DB	-43.2	79.8	4.2	3.9	-46.2	82.6	-48.6	84.8	-51.0	87.0	-54.0	89.8	(4)
(5)			WB	-43.1	65.3	3.9	2.1	-45.9	66.9	-48.2	68.1	-50.4	69.3	-53.2	70.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.9	-4.7	-3.3	6.8	21.9	38.2	49.9	56.8	55.1	44.5	26.5	7.8	-2.7
	DBStd	24.74	13.89	12.80	11.68	9.48	5.60	5.57	4.24	4.24	6.80	9.96	13.00	13.76
	HDD50	9634	1696	1491	1340	842	369	70	2	9	187	728	1267	1633
	HDD65	14646	2161	1911	1805	1292	832	454	258	309	616	1193	1717	2098
	CDD50	465	0	0	0	0	2	66	211	165	21	0	0	0
	CDD65	2	0	0	0	0	0	0	2	0	0	0	0	0
	CDH74	84	0	0	0	0	1	12	62	9	0	0	0	0
	CDH80	7	0	0	0	0	0	1	6	0	0	0	0	0
Wind	WSAvg	5.7	5.7	5.6	6.0	6.3	6.1	6.0	5.3	5.1	5.1	5.6	6.1	6.1
Precipitation	PrecAvg	21.0	0.6	0.5	0.7	1.0	1.3	1.9	2.7	4.1	3.0	2.4	2.0	0.9
	PrecMax	29.3	2.4	1.4	2.6	6.1	4.0	5.0	11.7	9.4	6.8	5.9	5.6	4.5
	PrecMin	12.4	0.0	0.0	0.0	0.1	0.1	0.1	0.1	1.3	0.4	0.4	0.0	0.0
	PrecStd	5.0	0.5	0.4	0.6	1.2	0.9	1.2	2.2	2.2	1.5	1.3	1.5	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	32.0	35.4	46.6	66.2	75.3	80.5	75.2	64.5	50.0	37.8	32.1
		MCWB	28.6	28.4	29.8	37.2	50.1	57.6	63.1	61.5	53.4	41.4	34.3	29.8
	2%	DB	27.8	27.1	31.9	41.1	58.8	70.1	75.1	71.4	60.5	45.6	35.3	28.0
		MCWB	25.2	24.2	27.7	33.8	46.8	55.9	61.4	58.8	51.9	39.9	32.5	26.0
	5%	DB	21.9	22.5	28.7	37.7	53.7	65.9	70.6	67.6	57.1	42.5	32.1	22.8
		MCWB	19.4	20.1	25.0	31.7	43.5	53.5	59.0	57.4	50.7	37.1	30.2	20.9
	10%	DB	17.3	15.9	25.5	35.2	48.6	61.1	66.4	63.8	54.4	39.5	28.2	18.0
		MCWB	15.1	13.6	22.2	30.4	40.9	51.6	57.5	56.0	49.6	35.2	26.5	15.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.4	29.1	30.9	37.6	51.3	59.9	65.5	63.5	57.4	44.1	35.1	30.4
		MCDB	31.7	31.5	33.9	44.4	64.2	71.7	76.2	71.9	61.8	47.2	37.0	31.0
	2%	WB	25.2	24.5	27.9	34.6	47.1	57.0	62.2	60.8	54.5	40.5	32.8	25.9
		MCDB	28.0	27.1	31.2	39.7	57.2	68.4	71.7	67.6	57.4	43.9	34.5	27.6
	5%	WB	19.5	20.1	25.3	32.6	44.2	54.6	60.2	58.6	52.7	38.0	30.1	20.9
		MCDB	21.8	22.5	28.7	36.7	52.2	63.7	68.7	64.4	55.4	41.3	31.9	22.8
	10%	WB	15.1	13.7	22.4	30.7	41.5	52.4	58.1	57.0	50.9	35.7	26.2	16.1
		MCDB	17.2	15.6	25.4	34.1	47.7	60.2	64.9	61.9	53.4	38.6	27.9	17.9
Mean Daily Temperature Range	5% DB	MDBR	22.6	23.9	25.7	20.9	17.3	20.1	18.1	17.1	17.8	18.7	21.5	20.4
		MCDBR	18.3	16.1	23.6	21.5	26.9	29.6	27.9	26.0	21.8	18.9	12.1	16.2
		MCWBR	16.1	14.6	19.6	15.1	16.0	16.3	15.5	15.2	14.1	13.8	10.6	15.0
	5% WB	MCDBR	17.4	15.4	20.4	18.1	25.0	26.2	24.4	21.5	16.5	16.5	11.2	16.4
		MCWBR	16.0	14.2	17.3	12.8	15.4	15.3	14.3	13.2	11.4	12.9	10.3	15.2
Clear-Sky Solar Irradiance	taub	0.229	0.275	0.334	0.428	0.461	0.428	0.483	0.437	0.349	0.316	0.258	0.214	
	taud	2.056	2.079	1.957	1.754	1.854	2.193	2.054	2.177	2.405	2.197	2.037	1.964	
	Ebn at Noon	174	226	243	236	240	255	234	237	246	207	152	127	
	Edh at Noon	14	25	39	59	57	42	47	38	26	22	14	10	
All-Sky Solar Radiation	RadAvg	137	432	960	1513	1670	1657	1498	1136	809	461	175	73	
	RadStd	12	31	60	106	157	182	180	143	107	52	23	7	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (1 neighbor)	N/A	N/A	-2.90	+1.82	+1.17	+1.03	N/A	N/A	+59	N/A
	N/A	N/A	-2.90	+1.82	+1.17	+1.03	N/A	N/A	+59	N/A

Nomenclature: See separate page